

# UniConnect

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## Annotations

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## Aim

The primary aim of this project is to develop UniConnect, a centralized web platform designed to streamline daily communication and group coordination among students, delegates, and teachers. It provides a structured, academic-focused alternative to informal messaging applications by offering role-specific tools for interaction and resource sharing.

## Content summary

This document outlines the theoretical background, system architecture, and initial implementation details of the UniConnect platform. It details the core problem of fragmented academic communication, presents the database schema and API design, and describes the foundational features developed for the Minimum Viable Product (MVP).

## Methods

The project follows an iterative development methodology to ensure a user-centric design. The initial phase involved creating visual prototypes and defining detailed use cases for each academic role to establish clear system requirements. Technical implementation is divided into sprints, utilizing FastAPI for the backend, PostgreSQL for database management via Docker, and WebSockets for real-time chat capabilities. The testing approach includes ongoing API endpoint verification, backend unit testing, and planned User Acceptance Testing (UAT) to validate the core MVP functionalities against the initial use cases.

## Results so far

At the current stage, the foundational backend architecture and relational database schema have been successfully deployed using Docker containers. Core functionalities, including secure registration, and real-time messaging, are operational. Planned results for the upcoming development phases include finalizing the frontend integration for resource sharing, event scheduling, and the interactive polling system.

## Applicability

UniConnect is directly applicable to university cohorts, project teams, and educators seeking a unified environment for daily academic coordination. It benefits students by providing clear access to class updates and peer support, while enabling teachers and delegates to efficiently manage group announcements, schedules, and resources without relying on personal social media accounts.

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# 1 Introduction

## 1.1 Context and problems

In the modern academic environment, universities rely on robust Learning Management Systems (LMS) and official email networks for formal administration and heavy resource distribution. However, these traditional tools are often too rigid for the dynamic, day-to-day communication required within specific educational groups, such as a class cohort or a project team. Consequently, students, delegates, and teachers frequently resort to informal, third-party messaging applications (like WhatsApp or Discord) to ask quick questions, organize study sessions, or share immediate updates. This creates a distinct problem: academic communication becomes blurred with personal messaging, important group decisions or polls are difficult to track, and teachers lack a structured, professional channel for light, immediate interactions with their classes. We address this problem by providing a dedicated space for these daily interactions. This matters because improving the flow of communication and coordination within educational groups directly supports a more engaged, responsive, and organized academic community.

## 1.2 Aim of the project

The primary aim of this project is to develop UniConnect, a dedicated web-based coordination and communication platform designed specifically to facilitate daily interactions and group management among students, delegates, and teachers.

### **Project tasks/objectives**

To achieve this aim, the project is guided by the following specific tasks:

- To design a role-based user interface that clearly separates the needs and permissions of students, delegates, teachers, and administrators.
- To develop a secure backend system using FastAPI to handle user authentication, group management, and real-time data flow.
- To implement structured communication channels, including group messaging and targeted announcements, to replace informal social media chats.
- To build practical coordination tools, such as basic schedule sharing, event tracking, and polling mechanisms, enabling delegates and teachers to easily organize their cohorts.
- To evaluate the platform's core communication features through targeted testing with a pilot group of users to ensure functional reliability and ease of use.

### 1.3 Scope and constraints

The scope of this project is strictly defined to deliver a functional Minimum Viable Product (MVP) focused on group coordination. The MVP explicitly includes in-scope features such as secure role-based registration, group-specific messaging, event creation, basic polling, and lightweight file sharing for immediate class needs. Conversely, several features are explicitly out of scope: this platform does not aim to replace institutional LMS platforms, meaning complex grading systems, plagiarism detection, and assignment submissions are excluded. Additionally, native mobile applications (iOS/Android) and built-in video conferencing capabilities are out of scope for this iteration. The project is also constrained by a strict development timeline and the technical boundaries of deploying a containerized web application within a localized environment.

### 1.4 Methods

The approach to completing this work follows an iterative methodology focused on user needs and steady technical delivery. The process begins with the creation of visual prototypes to map out the user interface and establish a logical navigation flow for each role. Following the design phase, detailed use cases are defined to translate these visual concepts into concrete technical requirements. The actual development is then divided into manageable sprints, focusing first on establishing the database and API architecture, followed by frontend integration. Finally, the team plans to test the system continuously during development through API endpoint verification, concluding with User Acceptance Testing (UAT) based on the defined use cases to ensure the platform genuinely facilitates group communication.

### 1.5 Document structure

This document is organized into three main chapters that detail the development lifecycle of the UniConnect platform. Chapter 1 establishes the theoretical and conceptual background, exploring the core communication problem, defining the project's scope, and presenting the foundational system design through use cases and visual prototypes. Chapter 2 focuses on the technical implementation, outlining the system architecture, the database design, the chosen technology stack, and the sprint-based development methodology. Finally, Chapter 3 presents the testing and results, evaluating the platform's core features through targeted validation and discussing how the delivered Minimum Viable Product aligns with the initial project aims.

## 2 Chapter 1: Theoretical/Conceptual background

### 2.1 Domain Overview and Key Concepts

The domain of educational technology, particularly concerning Computer-Supported Collaborative Learning (CSCL), has undergone significant transformations over the past decade. Historically, academic coordination relied heavily on centralized, physical points of information, such as campus notice boards or formal administrative offices. With the digital shift, this paradigm evolved into asynchronous digital communication, primarily utilizing official university email networks and centralized Learning Management Systems (LMS). However, the modern academic ecosystem is inherently dynamic. A standard university cohort is not merely a group of individuals receiving top-down instruction; it is a complex, interactive community comprising distinct roles.

Students require immediate peer support and clarification, class delegates (or student representatives) need efficient mechanisms to gather consensus and broadcast urgent administrative updates, and teachers require lightweight channels to provide rapid guidance without the overhead of formal institutional announcements. Consequently, a critical concept in this domain is the necessity for "role-based digital coordination." Effective academic communication must acknowledge these differing responsibilities. When coordination tools fail to reflect the actual hierarchical and collaborative structure of a class, the result is fragmented communication, missed deadlines, and a general decrease in student engagement. The core concept driving this project is the creation of a unified digital environment that respects these role boundaries while facilitating seamless, real-time interaction.

### 2.2 Overview of Related Solutions and Approaches

To address the need for digital academic interaction, institutions and students currently rely on two disparate categories of software solutions, both of which present notable limitations for day-to-day coordination.

The first category comprises institutional Learning Management Systems, such as Moodle, Canvas, or Blackboard. These platforms are exceptionally robust for heavy-duty academic tasks: they excel at securely distributing large official files, managing complex grading rubrics, and processing formal assignment submissions. However, their architecture is inherently asynchronous and rigid. They are not designed for rapid, informal communication. Attempting to organize a spontaneous study session or ask a quick clarifying question about a syllabus via an LMS forum is often slow and unintuitive, leading to low adoption rates for daily coordination.

To compensate for the rigidity of LMS platforms, students and staff frequently adopt a second category of tools: commercial instant messaging and enterprise communication applications, including WhatsApp, Discord, Slack, and Microsoft Teams. While these platforms successfully solve the issue of real-time communication, they introduce new systemic problems. Applications like WhatsApp blur the critical boundary between personal life and academic responsibilities, leading to cognitive overload and "notification fatigue." Furthermore, informal groups on these platforms lack structural academic hierarchy; a teacher or a delegate possesses the same technical

permissions as any other user, making it difficult to pin vital announcements, conduct structured academic polls, or systematically organize resources. UniConnect is proposed as a bridge between these two extremes, offering the real-time agility of instant messaging paired with the structured, role-based organization necessary for an academic environment.

## 2.3 Key Terms and Definitions

To ensure clarity throughout this document, several key technical and conceptual terms must be explicitly defined within the context of the UniConnect project.

The concept of Role-Based Access Control (RBAC) is foundational to the platform's security and user experience. RBAC is a mechanism that restricts system access and feature availability based on the assigned role of the user—in this case, distinguishing between Administrators, Teachers, Delegates, and Students.

The platform's real-time communication relies on WebSockets, a distinct computer communications protocol providing full-duplex communication channels over a single TCP connection. Unlike standard HTTP requests, which require the client to constantly ask the server for updates, WebSockets allow the server to push new messages instantly to the client, enabling live chat functionalities.

From a project management perspective, this document focuses on the delivery of a Minimum Viable Product (MVP). An MVP is defined as the most stripped-down version of a product that can still be released to pilot users to solve their core problem—in this scenario, basic group coordination, messaging, and polling—without building complex, secondary features.

Finally, the deployment and architecture of the application utilize Containerization, specifically through Docker. Containerization refers to the packaging of software code with just the operating system libraries and dependencies required to run the code to create a single lightweight executable, ensuring the application runs consistently regardless of the host environment.

## 2.4 Requirements Perspective

For the proposed solution to successfully address the domain problem and serve its intended users, it must satisfy a specific set of both functional and non-functional requirements.

From a functional perspective, the system must definitively provide robust user authentication intertwined with the aforementioned RBAC system. A correct solution must ensure that upon logging in, a user's interface dynamically adapts to their role. For example, delegates and teachers must be granted functional permissions to create global announcements and initiate class-wide polls, whereas standard students must be restricted to responding to these prompts and utilizing the chat features. Furthermore, the system is required to provide real-time group messaging capabilities that automatically segregate users into their assigned class groups, preventing unauthorized access to external cohort discussions. The functional scope also demands a centralized repository module, allowing users to upload and retrieve lightweight academic resources directly within their group space, thereby eliminating the need to cross-reference external cloud storage links.

From a non-functional perspective, the system's architecture must prioritize security, usability, and maintainability. Security is a paramount need; a correct solution must employ robust cryptographic protocols, such as bcrypt for password hashing and JSON Web Tokens (JWT) for secure, stateless session management, ensuring that user data and academic conversations remain confidential. Usability requires that the graphical interface be intuitive, responsive, and accessible across various devices, acknowledging that students frequently switch between laptops and mobile web browsers. Finally, to meet the maintainability requirement, the application must be designed using a modular architecture. By containerizing the PostgreSQL database and the FastAPI backend, the solution ensures that future iterations, scaling efforts, or migrations to cloud servers can be executed with minimal friction and downtime.



### 3 Chapter 2: Implementation

## 4 Chapter 3: Testing/Results

## 5 Conclusions

## 6 References

## 7 Appendices